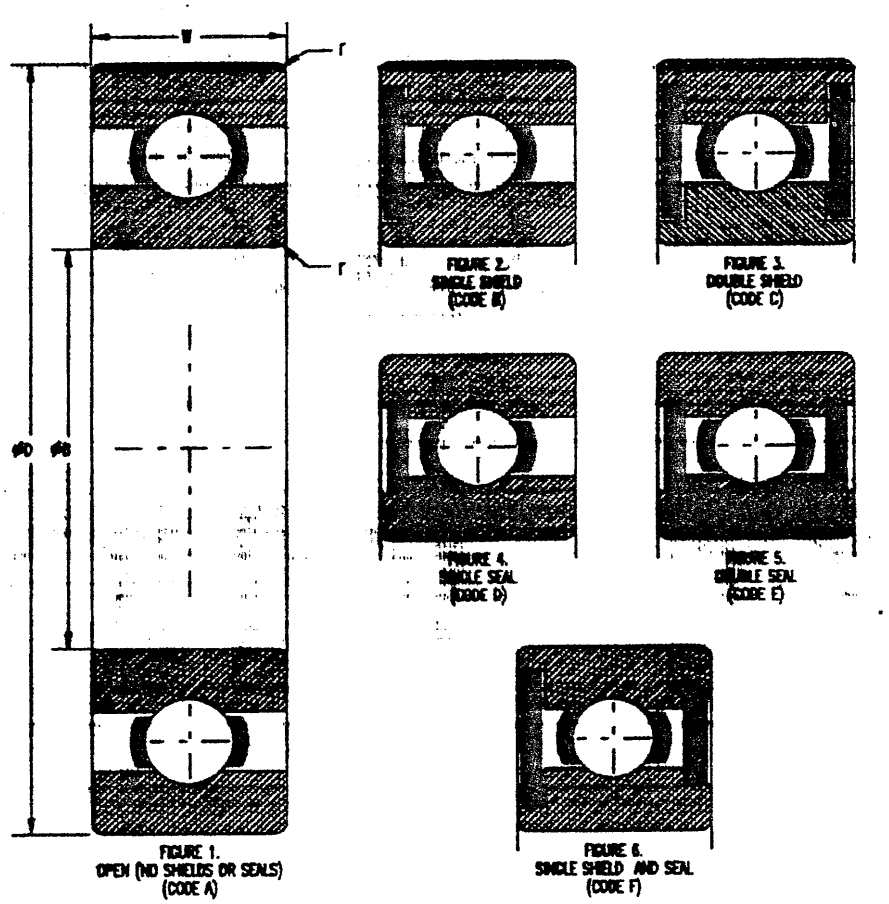
METRIC
FF-B-171/3
30 NOVEMBER 1993

FEDERAL SPECIFICATION SHEET

BEARING, BALL, ANNULAR, SINGLE ROW, RADIAL, NON-PILLING SLOT, ANSI/AFEMA DIMENSION SERIES 10.

The complete requirements for acquiring the bearings described herein shall consist of this document and the latest issue of



BEARING CONFIGURATIONS

AMSC/NA FSC 3110

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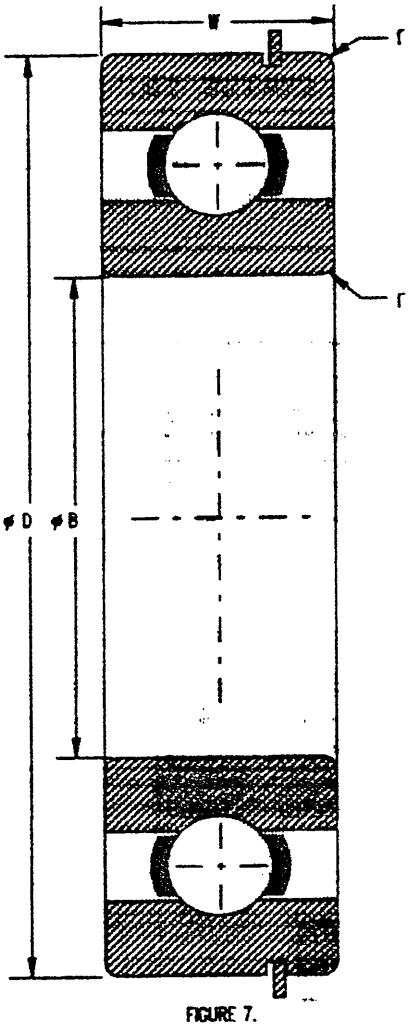


FIGURE 7.
OPEN SNAP RING
(CODE G)

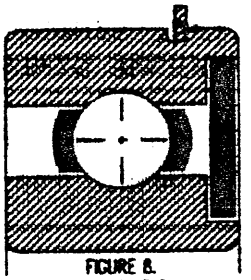


FIGURE 8.
SINGLE SHIELD-
ATTACHED ON SNAP
RING SIDE
(CODE H)

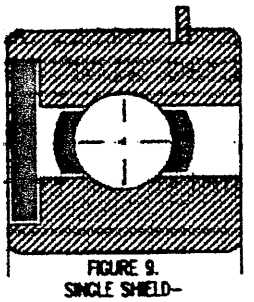


FIGURE 9.
SINGLE SHIELD-
ATTACHED ON SIDE
OPPOSITE SNAP RING
(CODE J)

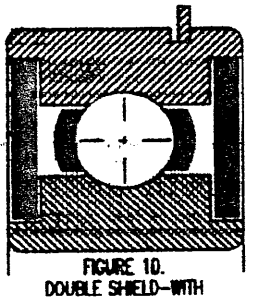


FIGURE 10.
DOUBLE SHIELD-WITH
SNAP RING
(CODE K)

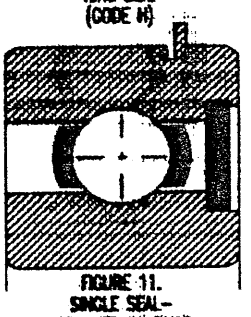


FIGURE 11.
SINGLE SEAL-
ATTACHED ON SNAP
RING SIDE
(CODE L)

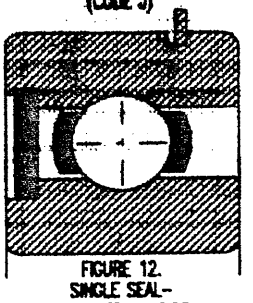


FIGURE 12.
SINGLE SEAL-
ATTACHED ON SIDE
OPPOSITE SNAP RING
(CODE M)

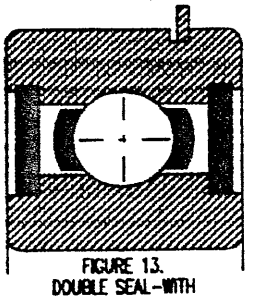


FIGURE 13.
DOUBLE SEAL-WITH
SNAP RING
(CODE N)

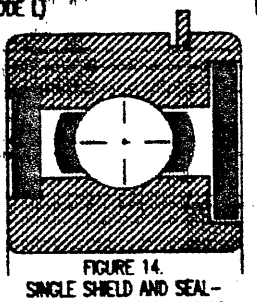


FIGURE 14.
SINGLE SHIELD AND SEAL-
SHIELD ON SNAP RING SIDE
(CODE P)

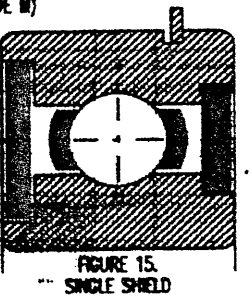


FIGURE 15.
SINGLE SHIELD
AND SEAL-SEAL
ON SNAP RING SIDE
(CODE R)

BEARING CONFIGURATIONS - (Continued)

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REQUIREMENTS:

1. **Boundary dimensions.** The boundary dimensions shall be specified in the part number by the size codes in Table I.

2. Cages. Cages shall be manufactured from any material unless otherwise specified. Cages shall be specified in the part number by the codes in table II.

3. **Shields** l seals, and snap rings. Shields and seals shall be fabricated from any material unless otherwise specified and shall be removable or nonremovable type. Seals shall be impervious contact type. Snap rings shall be fabricatd from ferrous material unless otherwise specified and shall have sufficient spring temper to set firmly in the ring grooves and shall be removable without the metal taking a permanent set. The snap ring and snap ring groove dimensions and their relative locations shall be in accordance with ANSI/AFBMA Standard 20. The number and positioning of shields, seals, and snap rings required shall be specified in the part number by the codes in table III.

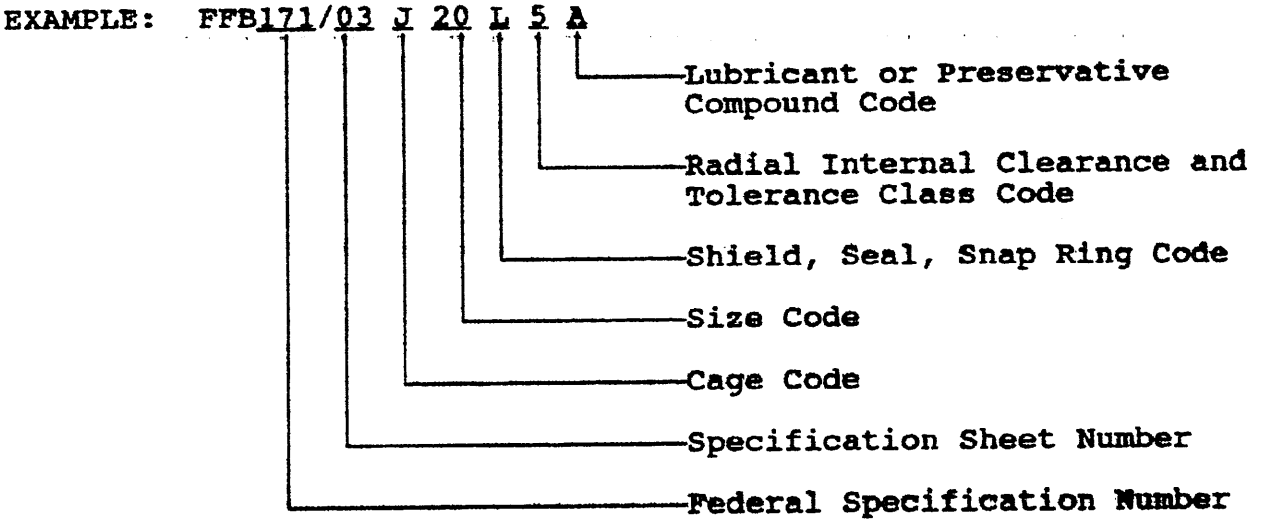
4. Radial Internal Clearance. The tolerance limits for radial internal clearance shall be in accordance with ANSI/AFBMA Standard 20 and shall be specified in the part number by the codes in table IV.

5. Tolerance Class. The tolerance limits for tolerance class ABEC-1 and ABEC-5 shall be in accordance with ANSI/AFBMA Standard 20 and shall be specified in the part number by the codes in table IV.

6. **Lubricant or Preservative Compound.** Preservative compound ‘ shall be applied to all open bearings (no shields or seals) and to single closure bearings (single shield or single seal) that have no additional lubricant requirement. Thus, for these bearings the bearing part number shall include the ‘C” suffix l For double closure or single closure bearings that require lubricant the bearing part number shall include the “A” or “B” suffix l When grease is required, the bearing void (the airspace between the inner and outer ring of the assembled bearing) shall be 25 to 50 percent filled. The lubricant or preservative compound shall be specified in the part number by the codes in table V.

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7. Part number. The part number consists of the letters FFB, the basic Federal specification number, the specification sheet number, and alpha or numeric codes for cage, size, shields, seals, and snap rings, radial internal clearance and tolerance class, and lubricant or preservative compound.



FFB171/03J201J5A indicates - Bearing, cage-pressed steel or molded plastic; bore 100mm; outside diameter 150mm; width 24mm; single seal attached on snap ring side; radial internal clearance symbol 2 (less than normal); tolerance class ABEC-5; grease in accordance with MIL-G-81322.

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TABLE I. Boundary dimensions.

Size Code 1/	OB Bore	OD Outside Diameter	W Width	Chamfer $r_s$min 2/	Shaft Shoulder Diameter Min 3/	C Dynamic load rating (pounds)
00	10	26	8	0.3	12.5	790
01	12	28	8	0.3	14.5	885
02	15	32	9	0.3	17.5	1040
03	17	35	10	0.3	19.5	1040
04	20	42	12	0.6	25	1630
05	25	47	12	0.6	30	1745
06	30	55	13	1.0	36	2295
07	35	62	14	1.0	41	2765
08	40	68	15	1.0	46	2900
09	45	75	16	1.0	51	3625
10	50	80	16	1.0	56	3770
11	55	90	18	1.1	62	4885
12	60	95	18	1.1	67	5085
13	65	100	18	1.1	72	5270
14	70	110	20	1.1	77	6570
15	75	115	20	1.1	82	6820
16	80	125	22	1.1	87	8225
17	85	130	22	1.1	92	8545
18	90	140	24	1.5	98.5	10045
19	95	145	24	1.5	103.5	10440
20	100	150	24	1.5	108.5	10365
21	105	160	26	2.0	115	12508
22	110	170	28	2.0	120	14155
24	120	180	28	2.0	130	14645
26	130	200	33	2.0	140	18350
28	140	210	33	2.0	150	18955
30	150	225	35	2.1	162	21680
32	160	240	38	2.1	172	23800
34	170	260	42	2.1	182	30165
36	180	280	46	2.1	192	31035

1/ Size codes in Table I are industry bore codes.
2/ The minimum bearing chamfer ($r_s$min) must clear the maximum shaft or housing fine! radius. This dimension does not control the shape of the bearing chamfer contour.
3/ For reference purposes only. Maximum shoulder heights shall be such as to permit-bearing removal pressure to be applied to the bearing inner ring. In cases where this requirement cannot be met, provisions shall be made for the removal of the bearing from the shaft without damage to t-he bearing or other parts.

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TABLE II. Codes for Cages.

Code	Cage
J	Pressed steel or molded plastic
M	Machined bronze or brass
T	Machined non-metallic (phenolic)

TABLE III. Codes for Shields, Seals, and Snap Rings.

Code	Shield, seal, and snap ring configuration
A	open (no shields or seals) - fig.1
B	single shield - fig.2
C	double shield - fig.3
D	single seal - fig.4
E	double seal - fig.5
F	single shield and seal - fig.6
G	open with snap ring - fig.7
H	single shield (attached on snap ring side) - fig.8
J	single shield (attached on side opposite snap ring)-fig.9
K	double shield with snap ring - fig.10
L	single seal (attached on snap ring side) - fig.11
M	single seal (attached on side opposite snap ring)-fig.12
N	double seal with snap ring - fig.13
P	single shield and seal (shield on snap ring side)-fig.14
R	single shield and seal (seal on snap ring side) - fig.15

TABLE IV. Codes for Radial Internal Clearance and ABEC tolerance.

Code	Radial Internal Clearance	Tolerance Class
1	Symbol 2 (less than normal)	ABEC-1
2	Symbol 0 (normal)	
3	Symbol 3 (greater than normal)	
4	Symbol 4 (greater than symbol 3)	
5	Symbol 2 (less than normal)	ABEC-5
6	Symbol 0 (normal)	
7	Symbol 3 (greater than normal)	
8	Symbol 4 (greater than symbol 3)	

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TABLE V. Codes for Lubricant and Preservative Compound.

Code	Lubricant or Preservative Compound
A	Grease in accordance with MIL-G-81322
B	Grease in accordance with DOD-G-24508
C	Preservative compound in accordance with MIL-P-197

MILITARY INTERESTS:

Military Coordinating Activity:
DLA - IS

Custodians:
Army - AT
Air Force - 11

Review Activities:
Army - AR, EA, MI
Navy - OS

User Activity:
Army - GL, ME

CIVIL AGENCY COORDINATING ACTIVITIES:
GSA-FSS

PREPARING ACTIVITY:
DLA-IS

(Project 3110-0$48-03)

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